

Date of Birth: 25 October 2001
Nationality: Spanish

www.linkedin.com/in/marina-crespo-aguirre 

<https://marinacrsp.github.io/> 

marinacrsp@gmail.com 

PROFILE

I am a curious, hard-working Biomedical Engineer, passionate about the fields of Machine Learning and Computer Vision and their application to solve problems relying at the intersection of Healthcare and Engineering, with specific focus on the Medical Imaging field.

Languages :

Spanish || English (C1) | French (C1) | German (A1)

Skills :

Signal & Image Processing · MATLAB · Python · Rstudio Machine Learning · Computer Vision

EDUCATION

BSc in Biomedical Engineering

Universidad Carlos III Madrid - Spain
2019 - 2023

University of Maryland - MD, USA
2021-2022

Relevant courses: Numerical Methods, Systems & Signals, Medical Image Processing, Computational Biology, Control Engineering (Honors), Instrumentation & Multimodality Imaging (Honors)

MSc in Medical Imaging

ETH Zürich
2023 - Fall 2025

Harvard Medical School - MA, USA
(Spring 2025)

Relevant courses: Introduction to Machine Learning, Deep Learning, Image Analysis & Computer Vision, Mobile health & Activity Monitoring, Medical Image Analysis

AWARDS

University Studies Grant

Fundación Emilio Botín - Santander

- Call September 2019
- Call September 2022

Post-Graduate Fellowship

La Caixa Foundation

Scholarship awarded to 100 Spanish & Portuguese graduate students to pursue post-graduate studies abroad.
(Call June 2023)

CONFERENCES & PAPERS

Physical & Engineering Sciences in Medicine (June 2023)

"Evaluation and assessment of clique arrangements for the estimation of omnipolar electrograms in high density electrode arrays"

doi: [10.1007/s13246-023-01287-8](https://doi.org/10.1007/s13246-023-01287-8)

Annual International Conference of the IEEE Engineering in Medicine and Biology Society (July 2023)

"Assessment of the Interelectrode Distance Effect over the Omnipole with High Multielectrode Arrays"

Annual Congress of the Biomedical Engineering Society (November 2022)

"Estudio Comparativo con Señales Epicárdicas de las Limitaciones del Omnipolo con Electrodo de Alta Densidad"

EXPERIENCE

Undergraduate researcher - University of Maryland

January - May (2022)

University of Maryland, College Park, USA

Design of a bio-inspired soft-robotic model of a marine turtle at BAM lab.

Skills : Solidworks, 3D printing, soft robotics, fluid mechanics

Undergraduate researcher - Bio-ITACA, UPV

May (2022) - September (2023)

Universidad Politécnica Valencia, Spain

Research on the development of novel biomarkers for the electrophysiologic characterization of cardiac substrate.

Byproducts from internship:

- [Journal Article](#)
- Oral and Poster Presentations at National and [International conferences](#)

Skills : Signal processing, MATLAB, Rstudio

Graduate researcher - Paul Scherrer Institute

February - June (2024)

Villigen, Switzerland

Master research internship in collaboration with ETH Zürich, on the investigation of non-invasive techniques for eye tracking, with application in proton therapy of ocular tumours.

Skills : Image analysis & processing, Eye Modelling, Computer Vision

Graduate researcher - CVL ETH

September 2024 - March 2025

Computer Vision Lab - ETH, Zürich, Switzerland

Master research internship on Deep Learning models (i.e. Neural Implicit Fields) for the reconstruction of MR images.

Skills : Python, Deep learning, Computer Vision, MRI reconstruction, Neural Implicit Fields

Visiting Student - Harvard Medical School

April 2025 - October 2025

Athinoula A. Martinos Lab for Biomedical Imaging - Boston, MA, USA

Master Thesis project on Deep Learning models (i.e. Diffusion Models) for the generation of synthetic Ex-vivo MRI scans from dissection photographs.

Skills : Python, Pytorch, Pytorch Lightning, Deep learning, Github, Diffusion Models